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ARTICLE

NiFe saponite as a new anode material for high-performance lithium-ion batteries†

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NiFe saponite (NF-SAP) with Na⁺ pillaring was investigated for the first time, as a new anode material for high-performance lithium-ion batteries (LIBs). In this material, non-metallic element Si and metallic elements Fe, Ni all act as active components (providing capacity) and undergo valence state changes during electrochemical progress. Si changes between Si²⁺ and Si⁴⁺; while Fe, Ni transform between nearly metallic state and Fe^{x+}, Ni^{y+} (0 < x < 3, 0 < y < 2). At full lithiation state, there are Ni–Ni bonds appearing but only Fe–O bonds can be observed in NF-SAP. The NF-SAP/Li half cell delivers a gradually increasing specific capacity in the initial hundreds of cycles (maximum capacity of 815 mAh g^{−1} at the 350th cycle) and maintains 646 mAh g^{−1} after 1000 cycles at 500 mA g^{−1}. The pre-pillaring effect of Na⁺ is responsible for the good Li⁺ storage performance, which leads to the decrease of Li⁺ transporting hindrance and high contribution ratio of pseudocapacitance. Furthermore, a full cell of LiMn₂O₄/NF-SAP was assembled, which shows an initial specific capacity of 104 mAh g^{−1} at 100 mA g^{−1} and maintains 68% after 50 cycles, demonstrating a bright application prospect of NF-SAP as promising anode material. This work discovers a new kind of anode for LIBs and provides a depth understanding of its energy-storage mechanism. Considering the easy preparation, low-cost and environment-friendly characteristics of NF-SAP, this kind of material is applicable to large-scale energy storage.

Introduction

The growing demand for energy has intensified the contradiction between the environment and humanity. Lithium-ion batteries (LIBs), as the most mainstream energy storage devices, have been investigated and commercialized for decades of years.^{1–3} The capability of the anode materials greatly affects the performance of the entire battery.^{4,5} Due to its high conductivity, low cost, and long cycle life, graphite has been widely used as the anode in commercial LIBs. However, because of its limited capacity (372 mAh g^{−1}), new anode materials have been developed to improve the energy density and power density for full batteries.^{6–9} Promising candidates mainly include silicon, tin, metal oxides, phosphorus, and their derivatives.^{11–14} Among them, the silicon-based anode has low discharge voltage and ultra-high specific capacity (4200 mAh g^{−1}). However, it is trapped in huge volume expansion/contraction (>400%), which prevents its practical application.^{15–17} The same problem plagues tin-based and metal oxide anodes. Although there are a number of ways to

modify them to achieve better performance,^{18–20} such as incorporation with carbon materials to relieve stress and improve electronic conductivity, there is still a lack of simple and cost-effective methods to completely solve such problems. A kind of low-stress anode materials with high structural stability may alleviate these dilemmas.

Saponites, as one type of layered materials have been extensively exploited as catalysts, catalyst supports and adsorbents.^{21–25} The host layer of saponites has a sandwich structure: two layers of Si–O tetrahedrons with one layer of Al–O(OH) octahedrons inside. The Si⁴⁺ can be replaced by trivalent metal ions (for instance, Al³⁺ or Fe³⁺), leading to an excess negative charge, which is neutralized by intercalated Na⁺, K⁺ or Ca²⁺ ions. The Al³⁺ in octahedrons can be substituted by other metal ions, such as Mg²⁺, Ni²⁺, Fe²⁺ or Fe³⁺. The interlamellar spacing of saponite is larger than or comparable with that of other two-dimensional materials, including layered metal hydroxide, hydrotalcite, MXene, graphite, transition metal sulfide, etc. Most of these layered materials have displayed outstanding performance in the energy storage field due to their ion intercalation property.^{26–31} Saponites with larger interlayer spacing may be more favorable for ions intercalation in electrochemical process, showing potential application prospects in energy storage field. However, saponites have been rarely developed as energy storage materials in the past.

In this work, we discovered the lithium-ion storage capability of Na⁺-pillared NiFe-saponite (NF-SAP) and explained the working mechanism of NF-SAP as an anode material. This

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is the first try to combine silicon-based anode with metal oxides delicately. The pre-pillaring of Na^+ makes the NF-SAP have a large interlamellar spacing of 11.6 Å to decrease the hindrance of Li^+ migration and increase the active surface area. Relied on combined valence state change of Si (Si^{2+} to Si^{4+}), Fe (Fe^0 to Fe^{x+}) and Ni (Ni^0 to Ni^{y+}) ($0 < x < 3$ and $0 < y < 2$), the NF-SAP/Li half-cell exhibits a high specific capacity of 815 mAh g^{-1} at 500 mA g^{-1} after 350 cycles without any structure optimization or material combination. Additionally, a LiMn_2O_4 /NF-SAP full cell was fabricated to study the potential application of saponite as LIBs anode. The full cell shows an initial specific capacity of 104 mAh g^{-1} and maintains 68% after 50 cycles. This work proves that the NF-SAP has great potential to be a new kind of anode for LIBs. There are reasons to believe NF-SAP will get better performance after a series of material optimization such as nanocrystallization or combination with highly-conductive materials.

Experimental

Reagents and materials

Analytically pure $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, NaHCO_3 , NaOH , Na_2SiO_3 , were purchased from Aladdin Co. Ltd. DodoChem Co. Ltd. supplied fluoroethylene carbonate (FEC), dimethyl carbonate (DMC), ethylene carbonate (EC), LiClO_4 , NaClO_4 and lithium foils. N-methyl-1,2-pyrrolidone (NMP) was acquired from Shanghai Ailei Chemical Co. Ltd. Shenzhen Kejing Co. Ltd provided Carbon ECP600JD (marked as ECP), polyvinylidene fluoride (PVDF) and LiMn_2O_4 . All the reagents were used as received without further purification.

Preparation of NF-SAP

The NF-SAP was prepared by a simple hydrothermal method. Typically, solutions A and B were prepared firstly, where solution A contains 90 mmol NaOH , 78 mmol NaHCO_3 , 25 mmol $\text{Na}_2\text{SiO}_3 \cdot 9\text{H}_2\text{O}$ in 100 ml de-ionized (DI) water; solution B includes 21.44 mmol $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$, 3.56 mmol $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ in 20 ml DI water. Solution B was added into solution A drop by drop under vigorous stirring. Then, the mixture was transferred to Teflon autoclaves and reacted at 180 °C for 48 h. The slurry was centrifuged with abundant DI water for 3 times and anhydrous alcohol for 1 time. Yellow solid product (Fig. S1, ESI†) was obtained after drying at 60 °C for 24 h.

Assembly of the coin-type half and full cells

Typically, 70 mg NF-SAP, 20 mg ECP, and 10 mg PVDF were ground for 2 min in a vibrating ball mill with four ZrO_2 (Φ 3 mm) balls and then 19 drops of NMP was put in for another 2 min milling. The slurry was coated on Cu foil with 150 μm thickness using a blade coating method. Li_2MnO_4 , ECP and PVDF were mixed into a mortar with a mass ratio of 18:1:1. NMP was dropped into the mixture accompanied by adequate grinding. When the slurry had appropriate viscosity, it was pasted on Al foil by a spreader with 150 μm thickness. Two kinds of wet electrode sheets were dried at 120 °C under vacuum overnight

and cut into circle sheets (Φ 12 mm). Half cells were assembled by using a piece of electrode sheet (LiMn_2O_4 or NF-SAP) as a working electrode and lithium foil as a counter electrode. The electrolyte was composed of 1 M LiClO_4 or NaClO_4 in a mixed solution of EC, DMC, and FEC (with a volume ratio of 5:5:1). For the full cell assembly, NF-SAP and LiMn_2O_4 electrodes played the role of negative and positive electrode, respectively, with an active mass ratio of 1:4. All cells were assembled under 800 psi for 5 seconds in an argon-filled glove box with water/oxygen content lower than 0.01 ppm.

Results and discussion

X-ray diffraction (XRD) pattern of the synthesized NF-SAP (Fig. 1a) matches well with those saponite materials in previous works.^{32, 33} A series of diffraction peaks at $2\theta = 7.6^\circ$, 19.9° , 28.9° , 34.9° , 54.1° , 62.0° , and 72.1° were attributed to (001), (100), (004), (110), (210), (300) and (220) planes, respectively. The d -spacing was calculated to be 11.6 Å based on the (001) lattice plane, corresponding to the Na^+ -pillared saponite phase.^{34, 35} The NF-SAP exhibits a reunited lamellar morphology in scanning electron microscopy (SEM) (Fig. 1b) and transmission electron microscope (TEM) images (Fig. 1c). The element energy disperse spectroscopy (EDS) mapping result (Fig. S2†) exhibits the characteristics of uniform distribution for Si, Ni, Fe and Na elements with contents of 49.62%, 40.17%, 5.36%, and 4.84%, respectively (inset of Fig. 1b). Inductively coupled plasma-atomic emission spectroscopy (ICP-AES) test shows the molar ratio of Fe/Ni is around 1/6, agree with the EDS result. High-resolution TEM image (Fig. 1d) shows clear lattice fringes with a distance of 11.6 Å, consistent with the interlamellar spacing obtained by the XRD pattern.

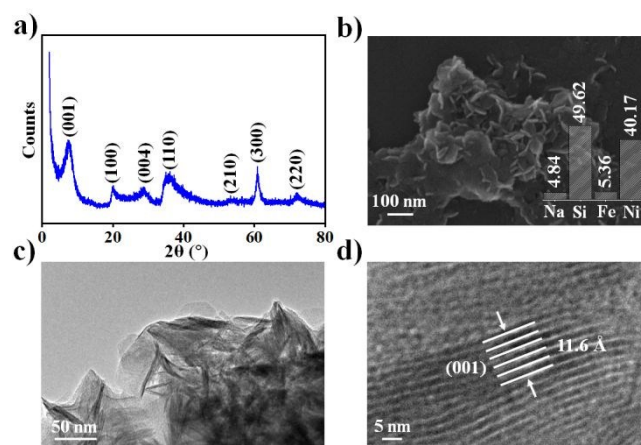
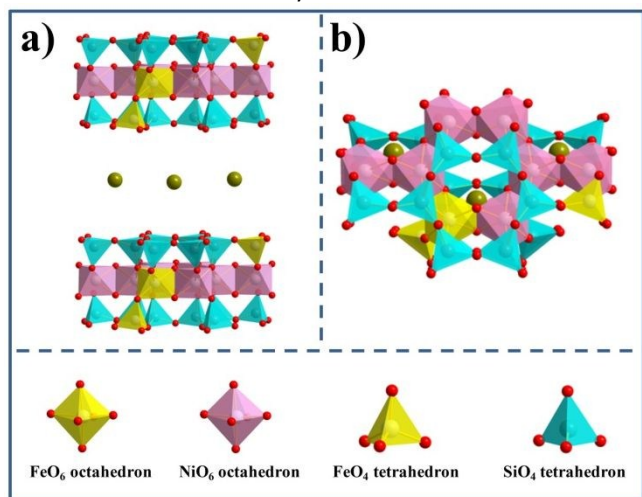


Fig. 1 (a) XRD pattern, (b) SEM image and EDS result and (c) TEM image of NF-SAP. (d) Lattice fringes of (001) crystal planes.

Fourier transform infrared (FTIR) spectra and complete attribution of the FTIR peaks are shown in Fig. S3 and Table S1 (ESI†). All absorption peaks are identified with those in previous works about the synthesis of saponites.³² To determine the chemical formula of NF-SAP, the amount of

crystal water is measured by thermogravimetry and derivative thermogravimetry (TG-DTG) analysis (Fig. S4†). A 15.4% weight loss appears in the temperature interval from room temperature to 900 °C coupled with a sharp DTG peak at approximately 100 °C deemed to the dehydration process.

Combined with the EDS, ICP-AES and TG-DTG results, the chemical formula of the as-synthesized NF-SAP is determined to be $[\text{Si}_{7.14}\text{Fe}_{0.86}]^{\text{t}}[\text{Ni}_{5.81}\text{Fe}_{0.14}]^{\text{o}}\text{O}_{20}(\text{OH})_4[\text{Na}_{0.81}]\cdot 10.1\text{H}_2\text{O}$ (the superscript "t" means tetrahedron while "o" represents octahedron). According to the previous understanding of saponite-like compounds, the layered structure of NF-SAP is shown in Scheme 1. The integrated structure is composed of sandwich-like metal oxide host layers and intercalated sodium ions. The host layers are constituted by two layers of SiO_4 tetrahedrons and one interlayer NiO_6 octahedrons, in which SiO_4 and NiO_6 were partially replaced by FeO_4 and FeO_6 , respectively. The oxygen atoms occupy the vertex positions of these octahedrons and tetrahedrons. The hydrogen atoms (not shown in Scheme 1 for clarity) are linked to those oxygen atoms that are not shared by octahedrons and tetrahedrons.



Scheme 1 The structural model diagram of NF-SAP: (a) side-view and (b) top-view. The interlamellar water is not shown for ease of understanding.

Fig. 2 displays the X-ray photoelectron spectroscopy (XPS) spectra of the pristine NF-SAP. Fig. 2a notably indicates the existence of Ni, Fe, Na, Si and O elements. The binding energy of Fe $2p_{3/2}$, Fe $2p_{1/2}$ are located at 711.3 and 723.0 eV, respectively (Fig. 2b), attributed to Fe^{3+} species.³⁶ In Fig. 2c, the O 1s peak can be divided into two parts: one with binding energy at 529.9 eV belongs to lattice oxygen and the other one at 531.2 eV is allocated to the oxygen of hydroxyl.³⁷ Fig. 2d exhibits the characteristic peak of tetravalent silicon (Si^{4+}) with the location of 101.1 eV. The peaks of Ni are well split into four parts, Ni $2p_{3/2}$ at 855.0 eV, Ni $2p_{1/2}$ at 873.0 eV and their satellites individually located at 860.8 and 878.8 eV (Fig. 2e), implying the presence of Ni^{2+} .³⁸ The Na 1s peak at 1071.0 eV indicates the existence of Na^+ in the interlayers of NF-SAP (Fig. 2f).

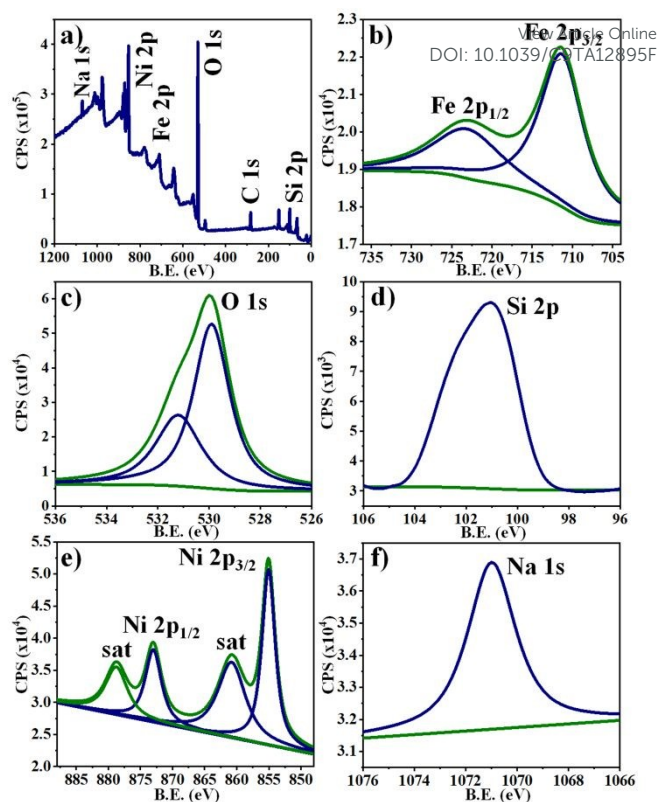


Fig. 2 XPS spectra of NF-SAP: (a) survey spectrum, (b) Fe $2p_{3/2}$, (c) O 1s, (d) Si $2p$, (e) Ni $2p_{1/2}$ and Ni $2p_{3/2}$, (f) Na 1s.

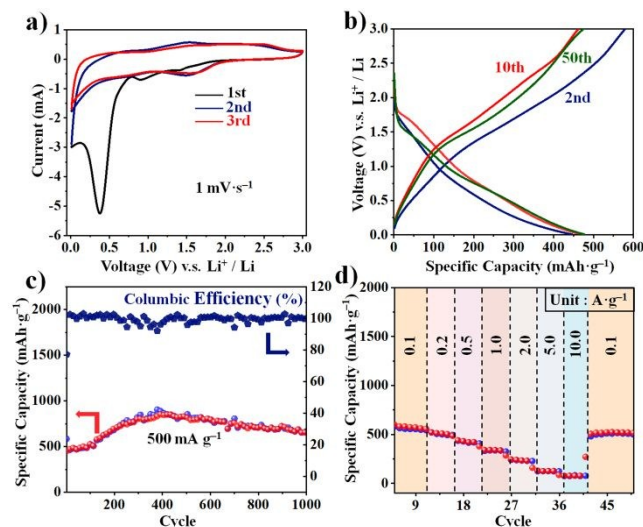


Fig. 3 (a) First three cycles of CV curves for the NF-SAP/Li half cell at a scanning rate of 1 mV s^{-1} . (b) Galvanostatic charge-discharge curves of the 2nd, 10th and 50th cycles for the NF-SAP/Li half cell. (c) Galvanostatic (dis)charging test at 500 mA g^{-1} , and (d) rate performance of the NF-SAP/Li half cell.

A NF-SAP/Li half cell was firstly used to study the electrochemical property of NF-SAP. The current collectors (Cu and Al foils) were proved to contribute nearly no capacity (Fig. S5†). From the CV curve (Fig. S6†) of the NF-SAP/Li half-cell, three oxidation peaks at 1.5 V, 2.0 V and 2.5 V along with two clear reduction peaks at 0.75 V and 1.6 V can be observed. As

shown in Fig 3a and Fig. S7[†], the CV and charging-discharging curves of the 1st and 2nd cycles have a significant difference, which is probably caused by the formation of solid electrolyte interface (SEI) film and activation of material (formation of Li_2O and Li_4SiO_4). The galvanostatic charge-discharge curves in Fig. 3b do not reveal platforms but slopes, suggesting no serious phase transformation. Fig. 3c displays the cycle performance of the NF-SAP/Li half cell at 500 mA g^{-1} . Different from normal phenomena, the specific capacity gradually increases with increasing the number of cycles. The coulombic efficiency hovers around 100% in these cycles, indicative of superior reversibility. The specific capacity creeps up to 815 mAh g^{-1} (350th cycle) from less than 500 mAh g^{-1} (original), disclosing long-time activation progress of NF-SAP, and maintains 646 mAh g^{-1} after 1000 cycles. The long activation time is probably related to the poor electrical conductivity of NiFe saponite. HRTEM images with lattice fringes of NF-SAP at 0.01 V and 3.0 V distinctly indicate that the layered structure is not destroyed in the electrochemical process (Fig. S8[†]).

The rate performance of the NF-SAP/Li half cell was evaluated at current densities from 0.1 to 10.0 A g^{-1} (Fig. 3d). When the current density is 0.1, 0.2, 0.5, 1.0, 2.0, 5.0 and 10.0 A g^{-1} , respectively, homologous specific capacity reaches 550, 495, 423, 331, 232, 119 and 80 mAh g^{-1} , respectively and it rapidly recovers to initial level when current density switches to 0.1 A g^{-1} from 10.0 A g^{-1} , displaying a good rate performance.

To understand the Li-ion storage mechanism of the NF-SAP anode material, XPS and *ex-situ* X-ray absorption near-edge structure analysis (*ex-situ* XANES) of Na, Fe, Ni, and Si were performed. The binding energy of Si 2p changes from 102.02 eV to 103.22 eV in the charging process and then migrates back to 102.00 eV after discharging (Fig. 4), illustrating the valence state of Si in NF-SAP changes between Si^{2+} and Si^{4+} in charging-discharging cycles. It was observed that the binding energy of Na invariably appears at 1071.72 eV without apparent intensity change (Fig. S9[†]). This result implies the Na^+ ions keep unmoved in the interlayer of NF-SAP, just acting like pillars to expand the interlaminar space, which is favorable to decrease the Li^+ transporting hindrance.

Besides the Si element, the Ni and Fe in NF-SAP also vary in valence state at different stages of charge (SOC). Fig. 5a and 5b show the Fe and Ni K-edge X-ray absorption near edge structure (XANES) spectra with Fe, Fe_2O_3 , Ni and NiO as references (Fig. S10[†] and S11[†] are complete X-ray absorption fine structure spectra of Fe and Ni). The pre-edge of Fe moves to higher energy from 0.01 V to 3.0 V, elucidating the oxidation of Fe. It is worth noting that the pre-edge of Fe at 2.2 V and 3.0 V nearly overlap, indicating the Fe has been completely oxidized to its maximum (less than +3) at 2.2 V. In addition, the Fe element at 0.01 V and 0.6 V almost have the same oxidation state as well. That is to say, the redox reaction of Fe happens in the voltage window of 0.6 V and 2.2 V. The valence state change of Ni has a similar trend at different SOC: the pre-edge of Ni shifts to higher energy region along with the rise of

voltage. What difference is that when Ni is at 2.2 V, it can still be oxidized to higher valence until 3.0 V (in Ni^{0+} state), the valence state of Ni at voltage under 1.4 V keeps almost unchanged (at Ni^{0+}). After full discharge from 3.0 to 0.01 V, the spectra for both Ni and Fe nearly overlap with the original spectra before charging (Fig. S12[†]), indicating high reversibility of the electrochemical reaction.

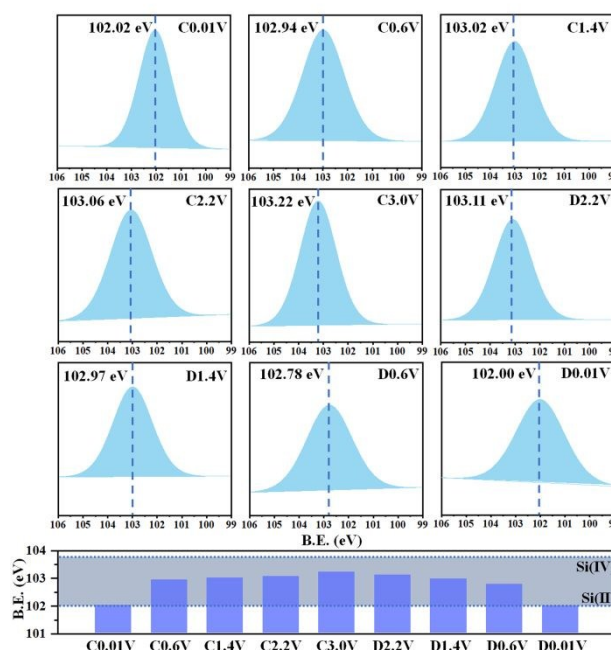


Fig. 4 XPS spectra and binding energy change of Si at different voltages in the NF-SAP/Li half cell (C and D mean charge and discharge, respectively).

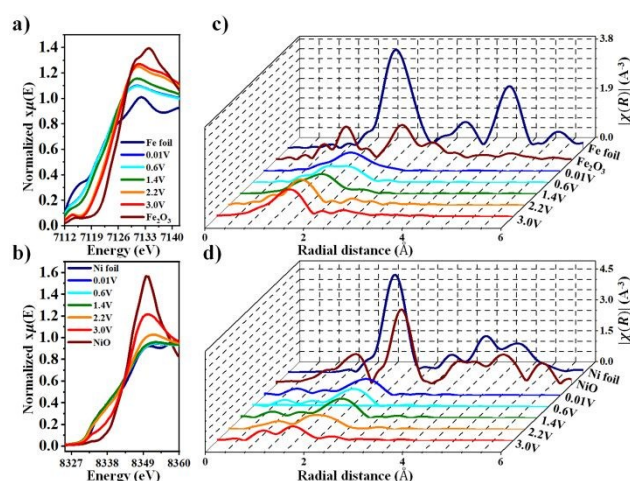


Fig. 5 XANES spectra at (a) Fe K-edge and (b) Ni K-edge at different SOC. (c) Fe and (d) Ni edge FT-EXAFS at different SOC.

Moreover, Fourier transformed extended X-ray absorption fine structure (FT-EXAFS) spectra offers detailed information about chemical bonds of Fe and Ni (Fig. 5c and 5d) to further understand the electrochemical behavior of NF-SAP. In Fig. 5c, no Fe–Fe bonds can be observed at all SOC because the Fe is evenly and sparsely dispersed in the NF-SAP host layers with

rather low content. The bond length of Fe–O gradually shortens when the cell is charged, proving elevated valence state of Fe in Fe–O bonds. In contrast, notable Ni–Ni bonds are observed below 1.4 V and Ni–O appears at voltages higher than 2.2 V. This result shows that metal nickel is produced at low voltages. The multivalent changes of Si, Ni and Fe lead to a high specific capacity of the NF-SAP anode material. Based on the above analysis, we proposed Li-storage equations of NF-SAP as follows (interlayer H₂O was ignored for clarity).

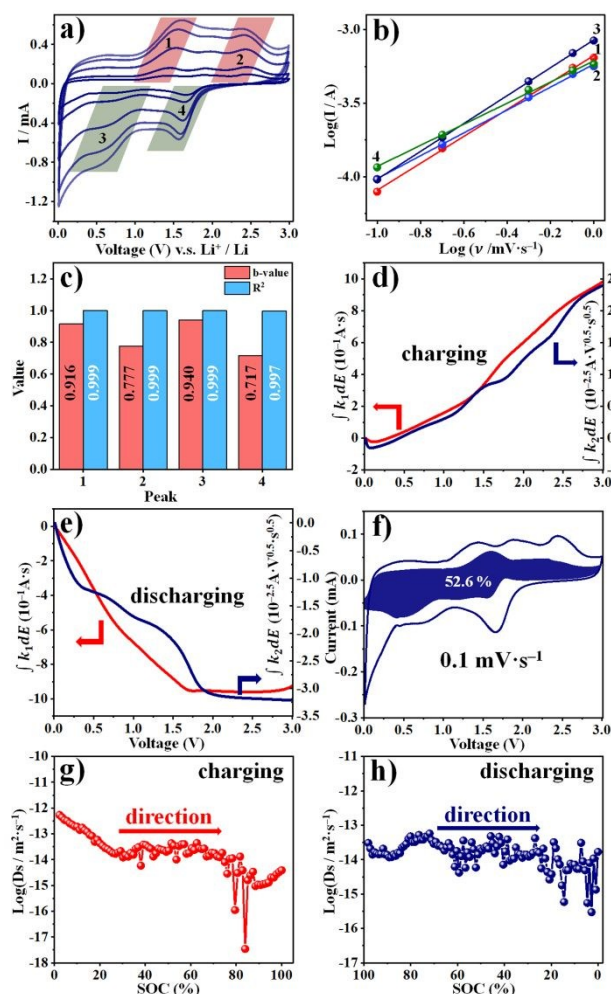
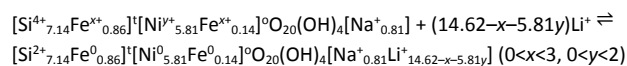


Fig. 6 (a) CV curves scanned at 0.1, 0.2, 0.5, 0.8 and 1.0 mV s^{−1}, (b) Plot of Log(*v*) and Log(*I*) depended on $i = k_1v + k_2v^{1/2}$, (c) *b*-value calculated from Fig. 4b, (d) $\int k_1dE$ and $\int k_2dE$ in the charging process, (e) $\int k_1dE$ and $\int k_2dE$ in the discharging process, (f) The ratio of pseudocapacitance in the specific capacity, (g) GITT test in the charging and (h) discharging process of NF-SAP/Li half cell.

CV curves at various scanning rates were used to study the dynamic characteristics of NF-SAP in electrochemical cycles (Fig. 6a). The formula $i = av^b$ is generally used to determine the dynamics of an electrochemical reaction, where *i* is peak current, *v* is scanning speed; *a* and *b* are kinetic parameters.^{39–41} The ordinary value of *b* is between 0.5 and 1.0:

b = 0.5 means diffusion-control while *b* = 1 represents surface control. Five CV curves at different scanning rates were plotted and four peaks were lined out in Fig. 6a; Log(*v*) and Log(*I*) at five scanning speeds of the four peaks in Fig. 6a were drawn in Fig. 6b and the data were linearly fitted. The slopes of four fitted lines in Fig. 6b are values of *b* and they are listed in Fig. 6c. The *b*-values of four peaks are 0.916, 0.777, 0.94 and 0.717, with related linearly dependent coefficients (*R*²) of 0.999, 0.999, 0.999 and 0.997, respectively. High *R*² means the credibility of the fitting and large *b*-value indicates the reaction in NF-SAP is mainly restricted by surface control (pseudocapacitance) with partial diffusion control (intercalation).

To understand the contribution of pseudocapacitance when NF-SAP works, Fig. 6d and 6e give out the change of $\int k_1dE$ and $\int k_2dE$ with voltage in the charging and discharging process, respectively. These two parts are further used in Fig. 6f relying on the modified formula $Q_F = \int i/vdE = \int k_1dE + \int k_2v^{-1/2}dE$. This amended formula can be used in the presence of polarization, where *i* is current, *v* is scanning speed, *E* is voltage, *k*₁ and *k*₂ are the same parameters in the formula $i = k_1v + k_2v^{1/2}$.⁴² According to this formula, the ratio of pseudocapacitance is calculated to be 52.6%. The high proportion of pseudocapacitance contribution may result in fast Li⁺ transport and better rate performance.

Galvanostatic intermittent titration technique (GITT) was used to determine the diffusion coefficient (*D*_s) of Li⁺ in charge-discharge cycles (see details in the ESI, Fig. S13[†]). As shown in Fig. 6g and 6h, the diffusion efficiency (*D*_s) in charging and discharging are in the range of 10^{−14}–10^{−12} m² s^{−1}. Such a high *D*_s is comparable with or superior to those of most anode materials (Table S2[†]), signifying fast Li⁺ ions transport in NF-SAP.^{43–50}

The fact that the NF-SAP is pillared by sodium ions makes it reasonable to consider whether this material can be used as electrodes in sodium-ion batteries (SIBs). To check this possibility, a NF-SAP/Na half cell with NaClO₄ electrolyte was assembled to detect its performance as a SIBs anode. Fig. S14[†] shows its initial capacity is only 280 mAh g^{−1} and the specific capacity decay to 64 mAh g^{−1} in the 38th cycle. Combining the fact that the XPS spectra of Na kept almost unchanged during lithiation/delithiation (Fig. S9[†]), these results mean the Na⁺ ions are difficult to migrate in the NF-SAP material, just locating in the interlayer space as pillars.

Based on the good Li⁺-storage performance of NF-SAP in half cells, a LiMn₂O₄/NF-SAP full cell (Fig. 7a) was fabricated to study the potential of NF-SAP for practical application considering that LiMn₂O₄ is a usual cathode material in LIBs. Fig. S15[†] shows the CV curve of LiMn₂O₄/Li half cell at 0.5 mV s^{−1} and constant current charging-discharging test at 100 mA g^{−1}. Take the specific capacity and potential window of LiMn₂O₄ and NF-SAP into consideration, the positive/negative electrodes mass ratio (P/N) of 4:1 and voltage window from 1 V to 4 V are chosen. The CV curve of the full cell (Fig. 7b) combines their characteristics of LiMn₂O₄ and NF-SAP. The two

pairs of peaks at 2.5/3.6 V and 1.8/2.5 V correspond to the (de)insertion of Li^+ into two kinds of tetrahedral voids in LiMn_2O_4 . Fig. 7c and 7d show the galvanostatic charging/discharging curve and cycling performance of the full cell. The $\text{LiMn}_2\text{O}_4/\text{NF-SAP}$ cell possesses an initial specific capacity of 104 mAh g^{-1} and maintains 68% after 50 cycles. Although the specific capacity has a fast attenuation, the coulombic efficiency holds around 100%. Amounts of factors can result in damping, such as P/N ratio, degree of overlap of cathode and anode, voltage window, etc. The nature of this full cell indicates its promising application prospect but needs further optimizing.

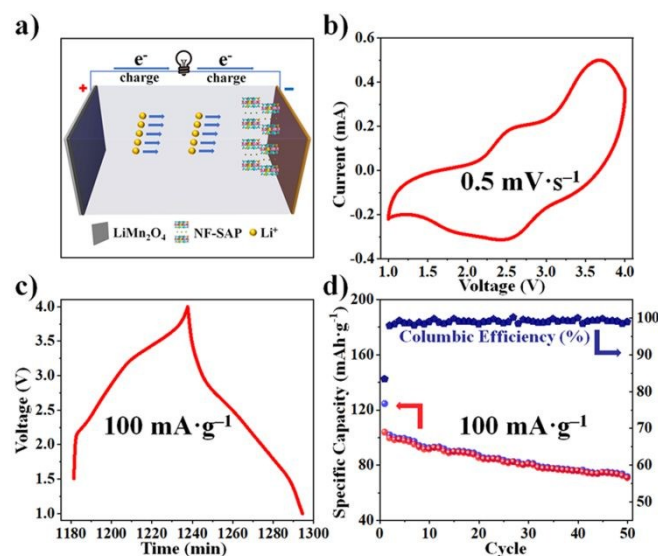


Fig. 7 (a) The scheme of $\text{LiMn}_2\text{O}_4/\text{NF-SAP}$ full cell. (b) CV curve of the full cell scanned at 0.5 mV s^{-1} . (c) The charging/discharging curves of the full cell at 100 mA g^{-1} . (d) Galvanostatic charging and discharging test of the full cell at 100 mA g^{-1} of $\text{LiMn}_2\text{O}_4/\text{NF-SAP}$ full cell with P/N ratio=4:1.

Conclusion

In summary, Na^+ -pillared NiFe-SAP with an interlayer distance of $11.6 \text{ \AA}$ was synthesized by a simple hydrothermal method, which exhibits outstanding Li^+ -storage performance as an anode of LIBs. The NF-SAP/Li half cell exhibits a high specific capacity of 815 mAh g^{-1} at 500 mA g^{-1} after 350 cycles and maintains 646 mAh g^{-1} after 1000 cycles. A $\text{LiMn}_2\text{O}_4/\text{NF-SAP}$ full cell is successfully constructed and runs at a voltage window of 1 to 4 V. The pre-pillaring effect of Na^+ may reduce the resistance of Li^+ migration reflected by large pseudocapacitance ratio and diffusion coefficient. The active components include non-metallic element Si and metallic elements Fe, Ni. All three of them have undergone valence state changes during electrochemical progress. Si changes between Si^{2+} and Si^{4+} ; while Fe, Ni switch between metallic state and Fe^{x+} and Ni^{y+} ($0 < x < 3$, $0 < y < 2$). This work discovers that saponite can be used as a promising anode for LIBs. In view of the ease of preparation, exchangeability of interlayer ions and

low-cost characteristics of saponite, we believe that saponite may be potentially used in broad energy storage areas.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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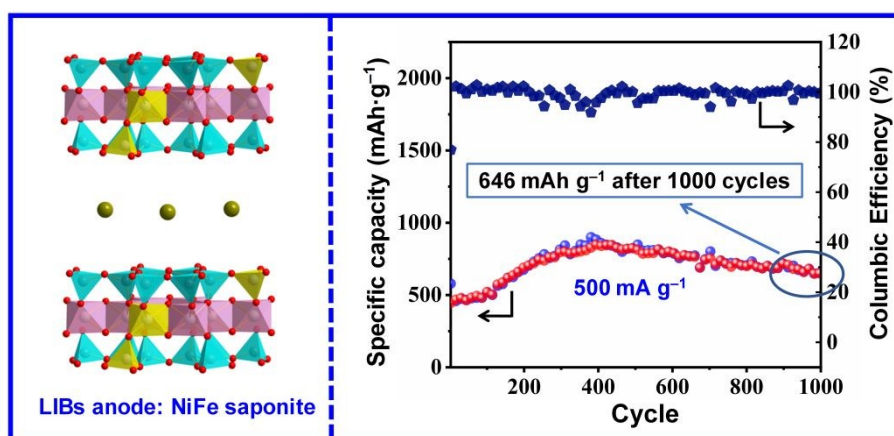
Table of Contents

NiFe saponite as a new anode material for high-performance lithium-ion batteries

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NiFe saponite was discovered for the first time as a new anode material for high-performance lithium-ion batteries, which delivered a maximum capacity of 815 mAh g^{-1} and maintained 646 mAh g^{-1} after 1000 cycles with charge/discharge density of 500 mA g^{-1} .